

AMENDMENT NO. 2 MARCH 1981
TO
IS : 2569 - 1978 SPECIFICATION FOR
MALATHION WATER DISPERSIBLE POWDER
CONCENTRATES
(Second Revision)

Alterations

[Page 4, Table 1, Sl No. (ii) and (iv)] — Substitute the following for the existing matter:

Sl No.	CHARACTERISTIC	REQUIREMENT	METHOD OF TEST, REF TO		
			CI No. and Appendix of this Standard	Appendix of IS : 1832-1978*	CI No. of IS : 6940-1973†
(1)	(2)	(3)	(4)	(5)	(6)
ii)	Malathion content after accelerated storage for 20 h at 90°C	Shall not be less than 88 percent of the initial value	5.2	—	—
iv)	Suspensibility after accelerated storage for 20 h at 90°C, percent by mass, Min	50	A	—	—

(Page 5, clauses 2.2.1 and 2.2.1.1) — Substitute the following for the existing matter:

‘2.2.1 Malathion Content — When determined by the methods prescribed in Appendix A of IS : 1832-1978*, the observed malathion content shall not differ from the declared nominal value by more than appropriate tolerance indicated below:

Nominal Value, Percent

Up to 9
Above 9 and below 50
50 and above

Tolerance

+ 10	} Percent of the nominal value
— 5	
± 5	
+ 5	
— 3	

2.2.1.1 The actual value of the malathion content in the formulations shall be calculated to the second decimal place and then rounded off to first decimal place before applying the tolerances given in 2.2.1.

2.2.1.2 The average content of all samples taken shall not be lower than the nominal content.'

(Page 6, clause 5.1.2) — Delete.

Addenda

(Page 6, clause 5.1.1) — Add the following new clause after 5.1.1:

5.2 Accelerated Storage Treatment

5.2.1 Weigh 20 g of the powder into a 100 ml wide mouthed glass bottle fitted with a vinyl-plastic-lined screw cap. Place in a forced draught oven maintained at $90^{\circ}\text{C} \pm 2^{\circ}\text{C}$. Allow the sample to remain in the oven for 20 hours and then cool to room temperature before removing the cap. After accelerated storage treatment, the sample should not be exposed to heat, bright-sunshine, or high atmospheric humidity. Proceed as per 5.1.1 for determination of malathion content.'

(Page 6, clause 5.2):

- a) Renumber the existing clause as 5.3.
- b) Add the following Appendix A after 5.3:

‘ APPENDIX A

SUSPENSIBILITY

[Table 1, Sl No. (iv)]

A-1. APPARATUS

A-1.1 Graduated Cylinder — 100-ml capacity having the 100-ml mark situated at $18.0\text{ cm} \pm 1.5\text{ cm}$ from the bottom.

A-1.2 Pipette — 25-ml capacity fitted with a device (for example a rubber stopper with a vent in the side) to permit its insertion into the 100-ml cylinder so that it is held with the top exactly at the 50-ml mark.

A-1.3 Water Bath — Water-bath maintained at $30 \pm 1^{\circ}\text{C}$ into which the 100-ml graduated cylinder can be immersed to the 100-ml mark. The bath must be free from any vibration caused by stirring motors or other equipment.

A-1.4 Glass Fibre Filter Paper

A-2. REAGENTS

A-2.1 Standard Hard Water — Dissolve 0.304 g of anhydrous calcium chloride and 0.139 g of magnesium chloride hexahydrate in distilled water and make up to one litre. This provides water with a hardness of 342 mg/l, calculated as calcium carbonate.

A-2.2 Potassium Bromide

A-3. PROCEDURE

A-3.1 Weigh accurately into a 100 ml beaker an amount of the treated sample (*see* 5.2) to form 100 ml of a suspension containing 2.5 percent of malathion (m/v). Add 50 ml of water at $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Stir the mixture with a glass rod by hand for 30 seconds, making no deliberate attempt to break up any lumps, and then immediately transfer the sample quantitatively to the 100-ml cylinder using additional water for the transfer. Add sufficient water at $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$ to make 100 ml of suspension. Stopper the cylinder and mix by inverting and righting it 30 times at the rate of approximately one cycle every two seconds. This operation should be carried out as smoothly as possible keeping the axis of rotation fixed. The cylinder must be thermally insulated from the hands to maintain the prescribed temperature of the suspension. Immerse the cylinder up to the 100-ml mark in the water-bath maintained at $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$. The preparation of the suspension, from the first addition of water to the placing of the cylinder in the constant-temperature bath, should be continuous operation and should be completed within 3 minutes. Allow the cylinder to stand for 30 minutes in the water-bath at $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$. During this period care should be taken that the bath and cylinder are free from vibrations.

NOTE — Should excessive flocculation occur during the test, the material shall be deemed unsatisfactory. Excessive flocculation in this context means either curdling up of the suspension in the entire column or the appearance of the transparent liquid accompanied by total or partial curdling up.

A-3.2 At the end of the 30-minute settling period, remove the cylinder from the water-bath, insert the specially fitted pipette so that the tip is exactly at the 50-ml mark and remove a 25-ml aliquot.

A-3.3 Transfer the 25-ml aliquot to a 100-ml beaker and add 2 g of potassium bromide. Allow to stand for about 5 minutes to permit the powder to coagulate. Place a 5.5-cm glass-fibre filter paper in a 5.5 cm Buchner funnel, insert the funnel in a 250-ml suction flask, and wet the paper with water. Transfer the coagulated 25-ml aliquot to the paper and wash with six 20-ml portions of water. Transfer the funnel to a dry 250-ml suction flask and extract the malathion with six 20-ml portions of acetone. Evaporate the acetone in the flask in a steam-bath to a volume of approximately 10 ml.

A-3.4 Transfer the residue quantitatively to a tared 50-ml beaker, using additional acetone. Evaporate the acetone at 60°C in a stream of dry air. Add two 5-ml portions of propan-2-ol during the evaporation to remove traces of water. Dry the sample in an oven at 55°C for 20 minutes and weigh as malathion.

A-3.5 Calculation

Suspensibility after accelerated storage, percent by mass = $m \times 160$

where

m = mass, in g, of malathion in 25 ml aliquot (see A-3.4)

(AFCD 6)